in the nuclei of leukocytes (pus cells) from discarded surgical bandages. The functions of nucleic acids were left undiscovered until early 1900.

7.1 Significance of Nucleic acids

- Nucleic acids are the molecular repositories (store houses / reserve bank) of genetic inheritance, i.e they have the ability to store and transmit information from one generation to the next.
- Ultimately, every macromolecule (proteins and RNAs) of the cell is the product of information that has been stored in the nucleotide sequences of the genes.
- Certain nucleotides like ribozymes have catalytic activities.
- Certain purine and pyrimidine analogs are used for treating cancer and AIDS.

7.2 Composition

Nucleic acids are very long, thread-like polymers, made up of a linear array of monomers called nucleotides held together by phosphodiester bridges. Nucleotides have three characteristic components (i) Base (ii) Sugar (iii) Phosphate group.

7.2.1 Common Bases of Nucleic acids

The bases of nucleic acids are heterocyclic, containing aromatic ring in their structures. They may be monocyclic pyrimidines or bicyclic purines.

i. Pyrimidine bases

- Pyrimidines are six-membered heterocyclic aromatic rings containing two nitrogen atoms (Figure 7.1) Pyrimidine ring is numbered in the clock-wise fashion. The common naturally occurring pyrimidines are cytosine, uracil, and thymine (5-methyluracil) (Figure 7.2). Cytosine and thymine are the pyrimidines typically found in DNA, whereas cytosine and uracil are common in RNA.

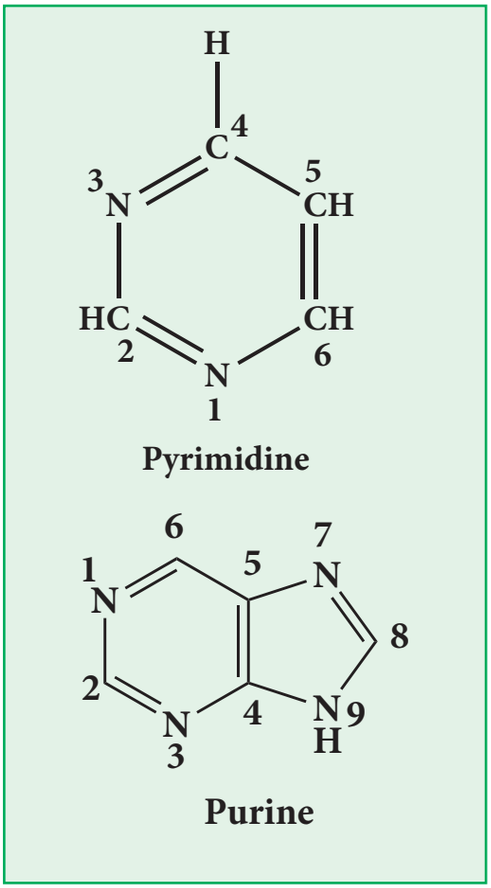

Figure 7.1. Structure of Purine and Pyrimidine

Properties of Pyrimidine bases

1. Pyrimidines are basic in nature. They are less soluble in water.

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